RAVENCRY — CIVIC REPORTING SYSTEM · ARCHITECTURE

USERS

- Citizens
- Operators

CITIZEN CHANNELS

- PWA — Vercel
- WhatsApp — Meta API
- Telegram — Bot API
- USSD — Feature phones
- SMS — No internet needed
- Appsmith — Operator dashboard

★ Feature phones — no internet required

⇄ SSE — PWA ↔ API Gateway live updates

EDGE LAYER

Internet / CDN (HTTPS) → Caddy reverse proxy (TLS termination · routing)

AWS/ALIBABA BOUNDARY · ECS · AURORA POSTGRESQL · POSTGRESQL RATE LIMITING TABLE · S3

DOCKER COMPOSE · ECS

APPLICATION LAYER

- API gateway — Rate limiting · routing
- Crucix engine — Core business logic
- LLM service — Qwen API bridge
- Whisper — Voice → text
- n8n — Workflows
- Telegram service — Bot polling
- SMS service — Africa's Talking
- USSD service — Session management

★ One-command deploy: docker compose up -d

Docker network — internal communication

AI PROCESSING PIPELINE

Voice report (Audio input) → Whisper (Transcription) → Qwen AI (Extract · Classify · Anti-fake; Hausa · Yoruba · Igbo · Pidgin · English)

★ Hausa · Yoruba · Igbo · Pidgin · English — code-switching

WORKFLOW · ORCHESTRATION

OSINT feeds (GDELT · ACLED · RSS) ↳ Crucix engine (Case management · alerts) → n8n (Broadcast triggers)

★ Two-way intelligence loop: Report → Alert → Sighting → Case update → New alert ↻

DATA LAYER

Aurora PostgreSQL + PostGIS
Cases · Persons · Alerts · Sightings
Subscribers · LGA coordinates
ST_DWithin geo queries

PostgreSQL rate limiting table
USSD sessions · 24h rate limit
Replaces DynamoDB ravencry-sessions

S3
Audio recordings
Photos
Attachments

★ ST_DWithin — subscribers within 160 km radius

★ 14 deliberate migrations — full referential integrity

NOTIFICATION LAYER

Broadcast engine
Fan-out to all channels

SMS
Africa's Talking

Telegram
Push to subscribers

Push notify
PWA subscribers

NAPTIP bridge
Authority handoff

EXTERNAL SERVICES — OUTSIDE AWS/ALIBABA BOUNDARY

WhatsApp API
Meta · Webhook

Telegram API
Webhook

Africa's Talking
SMS · USSD gateway

Qwen API
Alibaba Cloud

GDELT · ACLED
OSINT data feeds

Nigeria RSS
News aggregation

NAPTIP bridge
Gov integration

LEGEND

Citizen channels · notifications
Application services
AI services
Workflow · orchestration
AWS data stores
Operator tooling
External services
Synchronous request
External · async